IMDB Movie Data Analysis

Project Report

Exploratory analysis of the top 1,000 IMDB movies (2006-2016) with Python, pandas and Seaborn

1. Introduction

This project explores a dataset of the 1,000 highest-ranked movies on IMDB released between 2006 and 2016. The aim is to practise data cleaning, sorting, grouping, visualisation and correlation analysis while answering questions such as: Which movies are the best rated or highest grossing? Do longer movies perform better? How have ratings, revenue and output changed over the years? Which directors score highest?

2. Objectives

- Clean the data and summarise its main numeric features.
- Rank movies by rating, revenue, votes, Metascore and runtime.
- Compare long movies with shorter ones across four measures.
- Analyse trends by release year and the number of movies produced.
- Evaluate directors by average rating and test the link between rating and revenue.
- Label movies as Excellent, Good, Average or Bad based on rating.

3. Tools and Technologies

Tool	Purpose
Python	Programming language
pandas / NumPy	Data cleaning, filtering, grouping, conditional labelling
Matplotlib / Seaborn	Bar plots and scatter plot
Jupyter / Google Colab	Interactive development environment

4. Dataset

The file *7-IMDB-Movie-Data.csv* contains 1,000 rows and 12 columns after duplicates are dropped. Numeric columns are Rank, Year, Runtime (Minutes), Rating, Votes, Revenue (Millions) and Metascore. Text columns are Title, Genre, Description, Director and Actors. There are 644 unique directors.

Summary statistics (after cleaning)

Statistic	Runtime (min)	Rating	Votes	Revenue (M)	Metascore
Mean	113.2	6.72	169,808	71.97	55.21
Std. dev.	18.8	0.95	188,763	100.29	22.03
Min	66	1.9	61	0	0
Median	111	6.8	110,799	36.5	-
Max	191	9.0	1,791,916	936	-

5. Methodology

- **Load and de-duplicate:** read the CSV and apply `drop_duplicates()`.
- **Clean:** convert Revenue and Metascore to numeric, fill missing values with 0 and cast to integer.
- **Rank:** `sort_values(...).head(10)` for rating, revenue, Metascore, runtime and votes.
- **Compare by runtime:** split into the 5 longest movies and movies under 180 minutes, then compare mean rating, votes, revenue and Metascore.
- **Trend analysis:** Seaborn bar plots by Year for votes, revenue, rating, Metascore and movie count; `groupby` for yearly averages.
- **Directors:** mean rating per director, sorted to show the top and bottom 10.
- **Correlation:** Pearson correlation of Rating and Revenue with a scatter plot.
- **Categorise:** `np.select` into Excellent (8+), Good (6-8), Average (4-6) and Bad (under 4).

6. Findings

6.1 Top performers

Measure	Leader	Value
Rating	The Dark Knight (then Inception, Dangal at 8.8)	9.0
Revenue	Star Wars: The Force Awakens (then Avatar, Jurassic World)	936 M
Votes	The Dark Knight (then Inception, The Dark Knight Rises)	1,791,916
Metascore	Boyhood (then Moonlight at 99)	100
Runtime	Grindhouse (then The Hateful Eight)	191 min

Christopher Nolan films dominate the rating and votes lists. Critic favourites (Boyhood, Moonlight, Pan's Labyrinth) are not the same as the audience and box-office favourites.

6.2 Long versus short movies

The notebook compares the 5 longest movies (191 to 180 minutes) with movies under 180 minutes:

Measure	5 longest	Under 180 min
Average rating	7.68	6.72
Average votes	302,806	169,140
Average revenue (M)	49.56	83.11

Measure	5 longest	Under 180 min
Average Metascore	77.0	58.9

Long films score higher with audiences and critics and draw more votes, but earn less revenue on average. Because the long group has only five films, this should be treated as indicative rather than conclusive.

6.3 Trends by year

Year	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Movies	44	53	52	51	60	63	64	91	98	127	297
Avg rating	7.13	7.13	6.78	6.96	6.83	6.84	6.93	6.81	6.84	6.60	6.44

- The number of movies in the list rises steadily, from 44 in 2006 to 297 in 2016.
- Average rating is highest in 2007 (7.13) and 2006 (7.13) and lowest in 2016 (6.44).
- Average revenue per movie peaks in 2009, driven by Avatar (760 M), Transformers: Revenge of the Fallen, Harry Potter and the Half-Blood Prince and Up.
- The notebook concludes that more movies are being produced while revenue and audience engagement per movie decline.

6.4 Directors

Highest average ratings: Nitesh Tiwari (8.80), Christopher Nolan (8.68), Olivier Nakache and Makoto Shinkai (8.60). Lowest: Jason Friedberg (1.9), James Wong (2.7) and Shawn Burkett (2.7).

6.5 Rating and revenue

The correlation between rating and revenue is **0.22**, a weak positive relationship. A high rating helps financial performance only slightly.

7. Issues and Recommended Improvements

#	Issue	Recommended fix
1	Missing Revenue and Metascore are filled with 0. This produces zero-revenue and zero-score movies that pull down means (revenue 25th percentile is just 3 M) and distort rankings.	Keep as NaN (or use the median); pandas ignores NaN in means. Cast with the nullable Int64 type if integers are needed.
2	'Over 180 minutes' group is the top 5 by runtime (it includes three films at exactly 180). Variables data1 and data2 are reused for unrelated results.	Define the group with a clear threshold and a larger sample; use descriptive variable names.
3	Year bar plots show the mean per year, yet conclusions say 'most revenue in 2009' and 'most votes in 2012'. Revenue and votes are not totals.	State 'average revenue per movie', or plot sums with groupby().sum(). Also include movie counts, since 2016 has 297 films versus 44 in 2006.
4	'plt.figure(figsize=(100,10))' is called after the plot, so it creates an empty 10000x1000 figure.	Call plt.figure(figsize=(...)) before sb.barplot, with a sensible size such as (12, 5).
5	Director average rating includes directors with a single film, and mean rating is described as 'consistency'.	Require a minimum number of films and use standard deviation to measure consistency.

#	Issue	Recommended fix
6	A cell drops a column named 'average_rating' that is not in the dataset, and Revenue appears as NaN in the final output, which suggests cells were run out of order.	Remove the stray line and use Restart and Run All before saving.
7	Data is read from a Colab Drive path.	Use a relative path so the notebook runs anywhere.

8. Conclusion

The analysis highlights well-known favourites (The Dark Knight, Inception, Avatar, Star Wars: The Force Awakens), shows that critic scores and box office often diverge, and finds only a weak link between rating and revenue (0.22). Long films are rated higher but earn less on average, and the dataset grows heavily towards 2016. The main improvements are handling missing values properly instead of filling with zero, using larger and clearly defined comparison groups, and reporting totals or averages explicitly.

9. Future Work

- Analyse genres: average rating and revenue by genre, after splitting the Genre column.
- Study the most frequent actors and their average ratings.
- Plot a correlation heatmap for Rating, Votes, Revenue, Metascore and Runtime.
- Compare user Rating against critic Metascore.
- Build a simple regression model to predict revenue.
- Analyse the Excellent/Good/Average/Bad category distribution by year.